

Harthik Mallichetty

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WORK EXPERIENCE

Prolifics

February 2023 – July 2024

Associate Data Engineer | June 2023 – July 2024

Hyderabad, India

- Designed and deployed a TensorFlow-based sprint risk prediction service as a Flask REST API, giving 15+ engineering managers 1–2 weeks' advance visibility into at-risk sprints.
- Unified 250K+ JIRA and QA execution records through scalable SQL ETL pipelines, creating the trusted dataset that powered both predictive modeling and engineering analytics.
- Drove adoption across 200+ developers by integrating predictions into engineering dashboards, backed by documentation and onboarding across delivery teams.

Data Engineer Intern | February 2023 – May 2023

- Engineered SQL ETL pipelines that consolidated 100K+ JIRA and PostgreSQL engineering records into reusable analytics datasets, establishing the data foundation for delivery analytics.
- Standardized engineering KPIs for sprint health, defect trends, and delivery performance, creating a shared reporting framework across QA and delivery teams.
- Validated engineering datasets by identifying data quality issues across QA and project systems, improving the reliability of downstream analytics and reporting.

HeiDigi

March 2022 – August 2022

Data Analyst Intern

Hyderabad, India

- Built campaign measurement across 20+ marketing campaigns, defining KPI frameworks and automating recurring client reporting through reusable Python and SQL pipelines.
- Ran experimentation workflows and statistical analyses that identified high-performing audiences and creatives, driving 20% higher ROI and 18% higher CTR.

PROJECTS

Signal – Financial Intelligence (signal.harthik.dev)

December 2025

- Built a live platform turning hours of SEC filing research into seconds of natural-language Q&A across 70 S&P 500 companies via a 10-step RAG pipeline and a self-correcting LangGraph reasoning agent.
- RAG stack: 122,758 SEC filing chunks embedded with Voyage AI finance embeddings, pgvector HNSW retrieval, GPT-4o-mini generation, LangSmith observability on every query; deployed on GCP Cloud Run.

Quant Risk Desk (github.com/harthikerm/code-sentinel)

March 2026

- Automated financial risk desk providing real-time VaR, Sharpe, and Beta monitoring across cross-asset portfolios and market regimes — 21K records/day processed with 99th-percentile VaR confidence and zero violations. (Python, Pandas, SQL, Tableau)

EDUCATION

University of Texas at Dallas, Richardson, TX

May 2026

Master of Science in Business Analytics and Artificial Intelligence - Scholar With Distinction

GPA: 3.8/4.0

GITAM University, Hyderabad, India

April 2024

Bachelor of Technology in Computer Science and Business Systems

GPA: 3.7/4.0

SKILLS & INTERESTS

- Machine Learning & AI:** TensorFlow, PyTorch, Scikit-learn, Hugging Face, LangChain, LangGraph, OpenAI
- Data Engineering & Analytics:** ETL, Data Modeling, Airflow, dbt, Pandas, A/B Testing
- Languages & Databases:** Python, SQL, R, JavaScript, TypeScript, PostgreSQL, MySQL, pgvector, DuckDB
- Cloud & Backend:** GCP (Cloud Run, Vertex AI, Cloud SQL), Docker, FastAPI, Flask, Git, GitHub Actions
- Interests:** Golf, Fishing, Camping, Traveling, Cooking, Cricket, Soccer, Formula 1